

The mixed point spectrum of a matrix contraction on its closed unit ball

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Abstract

Let A be a contraction on a finite-dimensional complex normed space, and let $K_A f = f \circ A$ act on all continuous complex functions on its closed unit ball. We show that every unimodular eigenfunction factors through the contractive spectral projection onto the peripheral matrix eigenspaces. Consequently the unimodular point spectrum is exactly the multiplicative group generated by the peripheral matrix eigenvalues. This answers the mixed point-spectrum converse expressly left open in Kari Valentina Küster's 2015 thesis and included only in one direction in her 2016 *Pure Koopmanism* contribution. The argument is a short finite-dimensional application of the classical Jacobs–de Leeuw–Glicksberg mechanism. We give an elementary proof and collect the prior disk, peripheral and nilpotent cases, together with their full-spectrum consequences, in one table.

1 The question and the classification

Let $X = (\mathbb{C}^k, \|\cdot\|)$, where $k \geq 1$ and $\|\cdot\|$ is an arbitrary complex norm. For a matrix A with induced operator norm $\|A\| \leq 1$, put

$$U = \{x \in X : \|x\| \leq 1\}, \quad K_A : C(U) \longrightarrow C(U), \quad (K_A f)(x) = f(Ax).$$

Here $C(U)$ contains *all* continuous complex-valued functions and has the supremum norm. In particular U is compact, and $\|K_A\| = 1$ because constants attain the contraction bound. Write

$$\mathbb{T} = \{\zeta \in \mathbb{C} : |\zeta| = 1\}, \quad \mathbb{D} = \{\zeta \in \mathbb{C} : |\zeta| < 1\}.$$

The point spectrum $\sigma_p(T)$ consists of those ζ for which $Tf = \zeta f$ has a nonzero solution; $\sigma(T)$ denotes the Banach-operator spectrum.

Küster considered these exact matrix, norm, ball and observable hypotheses in [1, Section 3.1, printed pp. 35–43] and [2, Example 4, printed p. 321]. Her Theorem 3.1.14(i), printed p. 43 (PDF p. 52), proves

$$\mathbb{D} \cup \langle \sigma(A) \cap \mathbb{T} \rangle \subseteq \sigma_p(K_A)$$

when A has both a nonzero eigenvalue strictly inside the circle and a peripheral eigenvalue. The paragraph following that theorem expressly leaves the reverse inclusion open. Theorem 5(ii) of *Pure Koopmanism*, printed p. 321 (PDF p. 25), states the same inclusion. The completion here is the exclusion of additional unimodular eigenvalues.

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To state all cases, define

$$S = \sigma(A) \cap \mathbb{T}, \quad J = \{\alpha \in \sigma(A) : 0 < |\alpha| < 1\},$$

$$\Gamma = \langle S \rangle = \left\{ \prod_{\lambda \in S} \lambda^{m_\lambda} : m_\lambda \in \mathbb{Z} \right\}, \quad \langle \emptyset \rangle = \{1\}, \quad Z_A = \begin{cases} \{0\}, & A \text{ is singular,} \\ \emptyset, & A \text{ is invertible.} \end{cases}$$

Negative exponents are essential in the definition of the group. The bar on Γ below denotes closure in $\mathbb{C}$.

Theorem 1. *Under the hypotheses above,*

<i>condition</i>	$\sigma_p(K_A)$	$\sigma(K_A)$
$J \neq \emptyset$	$\mathbb{D} \cup \Gamma$	$\mathbb{D}$
$J = \emptyset$	$\Gamma \cup Z_A$	$\bar{\Gamma} \cup Z_A$.

In particular, in every case $\sigma_p(K_A) \cap \mathbb{T} = \Gamma$.

The all-peripheral point spectrum is prior work [1, Theorem 3.1.5, printed p. 38]; the nilpotent point spectrum is [1, Proposition 3.1.7, printed p. 39]; and the open disk from a nonzero stable matrix eigenvalue is [1, Theorem 3.1.13, printed pp. 41–42]. The stable-only equality is stated in [1, Theorem 3.1.14(ii), printed p. 43] and [2, Theorem 5(iii), printed p. 321]. The entire full-spectrum row for $J \neq \emptyset$ is already an immediate consequence of the prior open-disk inclusion, closedness of spectrum, and $\|K_A\| = 1$. The second full-spectrum row is a routine nilpotent direct-sum and peripheral-rotation consequence, proved below for completeness.

2 A projection compatible with the given norm

Lemma 2. *There is an A -invariant decomposition $X = E \oplus F$ and a commuting projection P onto E such that*

$$E = \bigoplus_{\lambda \in S} \ker(A - \lambda I), \quad A|_E = B, \quad A|_F = N, \quad N^n \longrightarrow 0, \quad \|P\| \leq 1.$$

Moreover $P(U) = V := U \cap E$, and B is a surjective isometry on E .

Proof. The bound $\|A^n\| \leq 1$ shows that $\sigma(A)$ lies in the closed disk and that every Jordan block associated with a unit-modulus eigenvalue has size one: a larger block has powers with polynomial growth. Take F to be the sum of the generalized eigenspaces associated with eigenvalues of modulus less than one. The resulting spectral projection P commutes with A , and the Jordan formula gives $N^n \rightarrow 0$ in operator norm.

We record the simultaneous recurrence needed to control P in the *original* norm. List the finitely many peripheral eigenvalues as $\lambda_1, \dots, \lambda_r$. Compactness of $\mathbb{T}^r$ supplies a strictly increasing sequence m_j such that $(\lambda_1^{m_j}, \dots, \lambda_r^{m_j})$ converges to a point of $\mathbb{T}^r$. For $n_j = m_{2j} - m_j$, we have $n_j \geq j$ and $\lambda_i^{n_j} \rightarrow 1$ for every i . If $r = 0$, simply take $n_j = j$. Since B is diagonalizable, this gives

$$B^{n_j} \longrightarrow I_E, \quad A^{n_j} \longrightarrow P.$$

Thus $\|P\| \leq 1$; this includes $E = \{0\}$, where $P = 0$. It follows that $P(U) \subseteq V$, while the reverse inclusion follows because P is the identity on E .

When $E \neq \{0\}$, B is invertible and $B^{n_j-1} \rightarrow B^{-1}$. Both B and B^{-1} have norm at most one in the restricted norm on E . Hence $\|Bx\| = \|x\|$, and $B(V) = V$. The zero-dimensional assertion has the same meaning on the singleton $V = \{0\}$. $\square$

Lemma 3. *If $f \in C(U)$ and $K_A f = \zeta f$ with $|\zeta| = 1$, then*

$$f = f \circ P.$$

Pullback $g \mapsto g \circ P$ identifies the unimodular eigenfunctions of K_B on $C(V)$ with those of K_A on $C(U)$.

Proof. For $x \in U$, both x and Px are in U . Commutation of A and P , and the eigenfunction equation, give for every $n \geq 0$

$$|f(x) - f(Px)| = |f(A^n x) - f(A^n Px)|.$$

The difference of the two arguments is $N^n(I - P)x$, which tends to zero uniformly for $x \in U$. Uniform continuity of f on the compact ball therefore makes the right side tend uniformly to zero. Thus $f = f \circ P$. Its restriction $g = f|_V$ satisfies $g(Bx) = \zeta g(x)$. Conversely that equation pulls back because $PA = BP$, and the pullback of a nonzero g is nonzero since $P(U) = V$. $\square$

3 The peripheral ball

We give the familiar all-peripheral calculation directly on $C(V)$. If $E = \{0\}$, the space V is a singleton and K_B is the identity on constants, so both its spectra are $\{1\} = \Gamma$. Suppose henceforth that $r = \dim E > 0$. Choose complex-linear coordinates $z_1, \dots, z_r$ diagonalizing B , so

$$z_i(Bx) = \lambda_i z_i(x), \quad |\lambda_i| = 1.$$

Every eigenvalue of K_B has modulus one, because $\|K_B f\|_\infty = \|f\|_\infty$. For nonnegative integer multi-indices p, q , the monomial $z^p \bar{z}^q$ is an eigenfunction with eigenvalue

$$\gamma = \prod_{i=1}^r \lambda_i^{p_i - q_i}.$$

It is nonzero on V , since V contains a neighborhood of zero in E . These eigenvalues exhaust Γ , proving $\Gamma \subseteq \sigma_p(K_B)$.

The polynomials in the coordinates and their conjugates are uniformly dense in $C(V)$ by the complex Stone–Weierstrass theorem: this algebra contains constants, separates points and is closed under conjugation. Let $\zeta \in \mathbb{T} \setminus \Gamma$, and suppose $K_B f = \zeta f$. Define

$$M_m = \frac{1}{m} \sum_{j=0}^{m-1} \zeta^{-j} K_B^j.$$

Since $B(V) = V$, K_B is an isometry and $\|M_m\| \leq 1$. On each monomial eigenfunction the multiplier is $m^{-1} \sum_{j=0}^{m-1} (\gamma/\zeta)^j$, which tends to zero. Consequently $M_m h \rightarrow 0$ for every such polynomial h . Given $\varepsilon > 0$, choose h with $\|f - h\|_\infty < \varepsilon$. As $M_m f = f$, we have

$$\|f\|_\infty \leq \varepsilon + \|M_m h\|_\infty,$$

and hence $f = 0$. No extra unimodular eigenvalue is possible. Together with Lemma 3 this proves

$$\sigma_p(K_A) \cap \mathbb{T} = \sigma_p(K_B) = \Gamma. \tag{1}$$

For later use we also have

$$\sigma(K_B) = \bar{\Gamma}. \tag{2}$$

Indeed K_B and its inverse both have norm one, so their Neumann-series resolvents exclude $|\zeta| > 1$ and $|\zeta| < 1$, respectively. Thus $\sigma(K_B) \subseteq \mathbb{T}$, while closedness gives $\bar{\Gamma} \subseteq \sigma(K_B)$. If Γ is infinite, its closure is $\mathbb{T}$. One elementary justification is that an infinite subgroup has nonidentity elements with arbitrarily small angular distance from 1; the successive powers of each such element form an arbitrarily fine net of the circle. If Γ is finite of order q , it is the group of q -th roots of unity and $B^q = I$. Hence $K_B^q = I$, and the factorization of $t^q - 1$ excludes every other spectral value. This proves (2).

4 Stable and nilpotent directions

4.1 A nonzero stable matrix eigenvalue

Suppose $\alpha \in J$, set $a = |\alpha| \in (0, 1)$, and choose a nonzero complex-linear functional ℓ with $\ell(Ax) = \alpha\ell(x)$. Such a left eigenfunctional exists for every matrix eigenvalue, including when A has nontrivial Jordan blocks. Let $R = \max_{x \in U} |\ell(x)| > 0$. For $0 < |\mu| < 1$, choose $\theta \in \mathbb{R}$ with $e^{i\theta} = \mu/|\mu|$, and put

$$s = \frac{\log |\mu| + i\theta}{\log a}, \quad f_\mu(x) = \begin{cases} \exp(s \log(|\ell(x)|/R)), & \ell(x) \neq 0, \\ 0, & \ell(x) = 0. \end{cases}$$

Here the logarithm in the function is the real logarithm of a positive number. Since $\Re s > 0$, the function is continuous across $\ker \ell$, nonzero, and $f_\mu(Ax) = a^s f_\mu(x) = \mu f_\mu(x)$. For $\mu = 0$, the nonzero continuous function

$$f_0(x) = \max\{0, |\ell(x)|/R - a\}$$

vanishes on $A(U)$, so $K_A f_0 = 0$. This reconstructs the prior open-disk inclusion by explicit radial eigenfunctions. The norm bound excludes all eigenvalues outside $\bar{\mathbb{D}}$; (1) excludes the extra boundary values. Therefore $\sigma_p(K_A) = \mathbb{D} \cup \Gamma$. Since spectrum is closed and contains the open disk, $\sigma(K_A) = \bar{\mathbb{D}}$, as already follows from the 2015 inclusion.

4.2 Only peripheral eigenvalues and zero

Suppose now that $J = \emptyset$. All matrix eigenvalues on F are zero, so $N^d = 0$ for some $d \geq 1$; if $F = \{0\}$, choose $d = 1$. The bounded projection

$$Q : C(U) \longrightarrow C(U), \quad Qf = f \circ P$$

commutes with K_A . Its range is isometrically identified with $C(V)$ by pullback, and the induced restriction of K_A is K_B . Its kernel is exactly the functions vanishing on V . For $f \in \ker Q$, the inclusion $A^d(U) \subseteq V$ gives $K_A^d f = 0$. Thus

$$C(U) = \text{ran } Q \oplus \ker Q$$

is a closed invariant direct sum, with a peripheral restriction and a nilpotent restriction.

When $F \neq \{0\}$, a nonzero linear functional vanishing on E restricts to a nonzero member of $\ker Q$. A nilpotent operator T on a nonzero space has zero as an eigenvalue: choose $h \neq 0$ and the smallest positive m with $T^m h = 0$, so $T^{m-1} h \neq 0$ lies in its kernel. Its spectrum is $\{0\}$, since for $\zeta \neq 0$ the finite geometric series in T/ζ gives a resolvent. When $F = \{0\}$, the nilpotent summand is absent. Spectra of a bounded operator on two closed invariant summands are the unions of the

restricted spectra, and the same holds for point spectra. Combining (1)–(2) proves the second row of Theorem 1, because $F \neq \{0\}$ here is equivalent to A being singular. This completes the proof.

In particular every nilpotent matrix, including $A = 0$, gives $\sigma_p(K_A) = \sigma(K_A) = \{0, 1\}$. The optional zero in the second row requires *singularity*, not that A be the zero matrix. In the first row zero is always an eigenvalue even if A is invertible: for example $A = \frac{1}{2}I$ maps U into a proper subset, and the function f_0 above lies in the Koopman kernel.

5 Classical mechanism and scope of the contribution

Lemma 3 is the elementary matrix-ball form of the classical Jacobs–de Leeuw–Glicksberg reversible projection, rather than a new general decomposition theorem. The connection can be checked explicitly. Let H be the matrix-norm closure of $\{A^n : n \geq 0\}$. It is a compact abelian semigroup. For every $f \in C(U)$, uniform continuity makes $L \mapsto f \circ L$ continuous from H to $C(U)$. Thus the Koopman powers have compact strong operator closure, which is also their weak operator closure. Their composition order agrees with multiplication because H is abelian.

The closure G_B of the powers of the surjective isometry B is a compact group. Recurrence and $N^n \rightarrow 0$ show that

$$G = \{DP : D \in G_B\} \subseteq H$$

is a group ideal with identity P . Every nonempty ideal of H intersects G , by multiplication by P ; its intersection with this group is then the whole group. Hence G is the minimal ideal, and P is its identity idempotent. The corresponding operator idempotent is exactly $Qf = f \circ P$. The classical JdLG theorem places every unimodular eigenvector in $\text{ran } Q$, giving the factorization; see [3, construction on printed pp. 308–309 and Theorem 16.33(a), printed p. 313]. Those references were checked in the author manuscript dated April 22, 2016, of the book published in 2015. The norm compatibility of P and the identification of Q above are the concrete applicability steps in this setting.

The theorem therefore answers a question expressly left open in the 2015 source and stated only as an inclusion in the 2016 source, by a short specialization of an existing mechanism. A bounded source audit located no earlier explicit completion in its examined scope; this is not a claim of historical firstness, exhaustive literature coverage, or continuous openness from 2015 to 2026. The accompanying priority note records the read scope and remaining coverage gaps.

All assertions concern the specified closed ball and the full continuous observable space. No invariant measure, holomorphic restriction, smoothness assumption, matrix diagonalizability assumption, or matrix invertibility assumption enters the theorem. Finite algebraic checks can test examples, projections and eigenfunction identities, but do not prove the infinite-dimensional spectral classification; the proof is the argument above.

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References

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